

04.1 — Deployment Runbook

For: Operators promoting code, fixtures, and customizations between dev / QA / prod. **Goal:** Zero-downtime, zero-data-loss promotion of changes through the env ladder. **Last updated:** 2026-08-29 **Pre-req reading:** 00.2 Architecture Overview, 01.1 Schema Reference, 04.4 Incident Response Plan, 05.1 Change Management Policy

Purpose

Promote code changes, schema changes, and master-data customizations from pberpdev → pberpqa → pberpprod safely, with backups before, smoke tests after, and a known rollback path.

This runbook covers:

- **Code changes** — haritha_hospital app code, custom scripts (Python / JS)
- **Schema changes** — new Custom Field, Property Setter, Print Format, Notification, Letter Head
- **Fixture changes** — additions to apps/haritha_hospital/fixtures/*.json
- **Master-data changes** — new Departments, Shift Types, Employees (run migrate_master_data.py)

It does NOT cover:

- **Fresh env provisioning** — see 04.3 Disaster Recovery Section “Full prod rebuild procedure”
 - **Infra changes** (container rebuild, nginx, compose) — different risk class; see 05.1 Change Management §“Change types”
 - **Hot bugfixes** — use the same flow but skip QA; see “Emergency hotfix” below
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Promotion ladder

pberpdev → pberpqa → pberpprod
(auto) (Venkat review) (Venkat approval + smoke)

Env	Approval required	Backup before	Smoke test
dev	solo (no approval)	optional (last successful slot = baseline)	mandatory before promote
qa	Venkat review	mandatory	mandatory
prod	Venkat approval + smoke plan	mandatory (Lesson #79 hardened)	mandatory + Venkat ack

Single-operator model — there’s no separate release manager. Whoever has write access drives the promotion.

Pre-flight checklist

Print this. Walk it. Tick every box before you touch a single container.

- [] 1. Last successful backup on target env verified (Lesson #79)
 - `ssh vijay@144.217.163.228 'ls -lt /home/vijay/backups/{env}/ | head -3'`
 - `ssh vijay@144.217.163.228 'cat /home/vijay/backups/logs/{env}_backup_*.log | tail -20'`
- [] 2. Offsite copy verified
 - `ssh venkat@135.125.196.35 'ls -lt /home/venkat/{env}_backups/ | head -3'`
 - SHA256 matches local bundle
- [] 3. All customizations exported as fixtures (Rule #9, Lesson #151)
 - `cd /home/vijay/frappeclaw/frappeclaw-data/workspace/projects/haritha-hospitals`
 - `git status` (no uncommitted fixture changes)
 - `ls apps/haritha_hospital/fixtures/` (must contain *.json for CF/PS/PF/N/LH)
- [] 4. hooks.py fixtures declaration present (Lesson #151)
 - `cat apps/haritha_hospital/haritha_hospital/hooks.py | grep -A 8 'fixtures = \['`
- [] 5. Customizations re-tagged with module='haritha_hospital' (Lesson #152)
 - `SELECT COUNT(*) FROM `tabCustom Field` WHERE module='haritha_hospital';`
 - same for Property Setter, Print Format, Notification
 - Letter Head: no module column – verify by owner/name (Lesson #155)
 - Skip Print Format/Notification with module IS NULL AND creation < 2024 (Lesson #156)
- [] 6. Tested on pberpdev (8-phase test plan per Lesson #48)
 - smoke → vendor tests → schema → CRUD → workflow → integration → regression → docs
- [] 7. Smoke test queued for post-deploy
 - write down: which 5 URLs to curl, which 3 doctypes to open in browser
 - share smoke plan in change record (see 05.1)
- [] 8. Rollback plan documented
 - "If smoke fails: (a) docker restart erp-{env}-backend-1 (if install-app), OR
 - (b) restore from backup at <path>, OR
 - (c) revert git commit + docker restart"
- [] 9. Venkat notified before prod change (Rule #13 from MEMORY)
 - Telegram message: "promoting {commit} to pberpprod at {time}. OK?"
- [] 10. MEMORY.md DB password verified fresh
 - `docker exec erp-{env}-db-1 printenv MYSQL_ROOT_PASSWORD`
 - if MEMORY's record differs, update MEMORY.md BEFORE proceeding (Lesson #154)
- [] 11. gunicorn restart step planned (Lesson #153)
 - docker restart erp-{env}-backend-1 ALWAYS after bench install-app
 - expect 30-60s downtime per env (acceptable, no data loss)
- [] 12. Off-hours if possible
 - prod deploys preferred 22:00-04:00 IST (low user activity)

→ emergency prod deploys: any time, but notify Venkat IMMEDIATELY
If ANY box unticked — STOP. Resolve before proceeding.

Dev → Prod promotion flow (step-by-step)

Phase 1 — Stage on dev

```
# 1. Branch from main (always branch, never commit direct to main)
ssh vijay@144.217.163.228
cd /home/vijay/frappeclaw/frappeclaw-data/workspace/projects/haritha-hospitals
git checkout main
git pull origin main
git checkout -b feature/my-change

# 2. Make changes
#   - code: apps/haritha_hospital/haritha_hospital/**/*.py
#   - JS: apps/haritha_hospital/haritha_hospital/public/js/*.js
#   - fixtures: apps/haritha_hospital/fixtures/*.json
#   - hooks.py: apps/haritha_hospital/haritha_hospital/hooks.py

# 3. Commit + push
git add -A
git commit -m "feat(haritha): <description>"
git push origin feature/my-change

# 4. On dev container – pull + apply
docker exec erp-dev-backend-1 bash -c '
  cd /home/frappe/frappe-bench/apps/haritha_hospital &&
  git pull origin feature/my-change'

# 5. If new fixture rows added via UI in dev – export + commit
docker exec erp-dev-backend-1 bash -c '
  cd /home/frappe/frappe-bench &&
  bench --site pberpdev.duckdns.org export-fixtures --app haritha_hospital'

# 6. If new Python package – install + RESTART backend (Lesson #153)
docker exec erp-dev-backend-1 bash -c '
  cd /home/frappe/frappe-bench &&
  pip install --user <package>' # avoid pip install -e for venv cleanliness
docker restart erp-dev-backend-1 # ← NEVER skip this

# 7. Smoke test on dev
curl -sS -o /dev/null -w "%{http_code}\n" https://pberpDEV.duckdns.org/ # expect 302 (red)
curl -sS -c /tmp/c.txt -X POST https://pberpDEV.duckdns.org/api/method/login \
  -H 'Content-Type: application/json' \
  -d '{"usr": "Administrator", "pwd": "admin123"}' | jq . # expect success
curl -sS -b /tmp/c.txt https://pberpDEV.duckdns.org/app/attendance | grep -i "500\|error"
```

Phase 2 — Promote to QA

```
# 1. Merge to main
git checkout main
```

```

git merge --no-ff feature/my-change
git push origin main

# 2. On QA container – pull + apply
docker exec erp-qa-backend-1 bash -c '
  cd /home/frappe/frappe-bench/apps/haritha_hospital &&
  git pull origin main'

# 3. Run migrate (only if hooks.py changed or new fixtures)
docker exec erp-qa-backend-1 bash -c '
  cd /home/frappe/frappe-bench &&
  bench --site pberpqa.duckdns.org migrate'

# 4. RESTART backend (Lesson #153)
docker restart erp-qa-backend-1
sleep 45 # let gunicorn warm up

# 5. Smoke test on QA (same commands as dev, replace URLs)
# 6. Notify Venkat for review

```

Phase 3 — Promote to prod

```

# 0. Pre-flight: confirm backup fresh (run /home/vijay/scripts/pberpprod_backup.sh manually)
ssh vijay@144.217.163.228 '/home/vijay/scripts/pberpprod_backup.sh' # takes ~3 min

# 1. Pull on prod container
docker exec erp-prod-backend-1 bash -c '
  cd /home/frappe/frappe-bench/apps/haritha_hospital &&
  git pull origin main'

# 2. Migrate (idempotent; safe to re-run)
docker exec erp-prod-backend-1 bash -c '
  cd /home/frappe/frappe-bench &&
  bench --site pberpprod.duckdns.org migrate'

# 3. CRITICAL: RESTART backend (Lesson #153 – gunicorn --preload sys.path freeze)
docker restart erp-prod-backend-1
sleep 45

# 4. Verify
curl -sS -o /dev/null -w "%{http_code}\n" https://pberpPROD.duckdns.org/ # expect 302
# 5. Login + smoke (replace password with MEMORY.md ADMIN_PASSWORD for prod)
curl -sS -c /tmp/c.txt -X POST https://pberpPROD.duckdns.org/api/method/login \
  -H 'Content-Type: application/json' \
  -d '{"usr": "Administrator", "pwd": "<from MEMORY>"}' | jq .
curl -sS -b /tmp/c.txt https://pberpPROD.duckdns.org/app/roster | grep -i "500\|error"

# 6. Notify Venkat: "prod deploy complete. {commit} live. smoke: {results}"

```

Re-tagging customizations (idempotent, scoped)

When a Custom Field / Property Setter is created in pberpprod via UI without module='haritha_hospital', it needs re-tagging so it exports correctly with the app.

Rule: re-tag ONLY what needs it. Don't bulk UPDATE.

-- Example: re-tag a Custom Field

```
UPDATE `tabCustom Field`  
SET module = 'haritha_hospital'  
WHERE name = 'Employee-custom_field_name'  
AND module != 'haritha_hospital';
```

-- Verify (must return 1 row):

```
SELECT name, module FROM `tabCustom Field`  
WHERE name = 'Employee-custom_field_name';
```

Decision rule (Lesson #156): if creation < 2024 AND module IS NULL, treat as stock. Don't re-tag.

Don't bulk-update with module IS NULL — you'll catch Frappe's old stock Print Formats / Notifications and ship them as your customizations.

After re-tag:

```
docker exec erp-{env}-backend-1 bash -c '  
  cd /home/frappe/frappe-bench &&  
  bench --site {site} export-fixtures --app haritha_hospital'  
# Commit + push fixture JSON
```

Common pitfalls

Pitfall	Symptom	Fix
Forgot fixtures = [...] in hooks.py (Lesson #151)	bench export-fixtures exits 0 but writes zero files	Add declaration per Lesson #151
Forgot docker restart erp-{env}-backend-1 after install-app (Lesson #153)	HTTP 500 on every request, ModuleNotFoundError in tabError Log	docker restart erp-{env}-backend-1
Used module = 'app' instead of module = 'haritha_hospital' in SQL	No rows updated	Lesson #152: v16 uses module column
Ran bench migrate without backup	Catastrophic if migration fails mid-way	Always backup first (Lesson #79)
Bulk re-tagged module IS NULL	Shipped Frappe stock Print Formats as customizations	Decision rule: skip creation < 2024 (Lesson #156)
Used docker-compose down instead of restart	Volumes preserved BUT all in-flight sessions killed (worse UX)	Always docker restart (Lesson #154)
Used docker-compose down -v	ALL DATA LOST	NEVER, unless wiping fresh (Lesson #154)
Confused bench restart with container restart	bench restart doesn't restart gunicorn; new Python packages invisible	Always docker restart for gunicorn-bound changes

Pitfall	Symptom	Fix
Pushed fixture JSON before commit on dev	Conflicting fixtures between envs	Commit on dev, then promote, never the reverse
Promoted without smoke test	Bug ships to prod before anyone notices	Pre-define smoke URLs in change record (see 05.1)
DB password drift (Lesson #154)	401 login on prod even though app code correct	docker exec erp-{env}-db-1 printenv MYSQL_ROOT_PASSWORD — verify before deploy

Rollback procedure

Pick the lightest rollback that fixes the issue. Don't restore from backup unless you must.

Level 1 — Restart only (fastest, safe)

For new Python package, gunicorn sys.path issues, transient errors:

```
docker restart erp-{env}-backend-1
sleep 45
curl -sS -o /dev/null -w "%{http_code}\n" https://pberpPROD.duckdns.org/
```

No data loss. 30-60s downtime. Reverses in-memory state only.

Level 2 — Revert code + restart (for bad commits)

```
# On container
docker exec erp-prod-backend-1 bash -c '
  cd /home/frappe/frappe-bench/apps/haritha_hospital &&
  git log --oneline -5' # find last-good commit

docker exec erp-prod-backend-1 bash -c '
  cd /home/vijay/frappeclaw/frappeclaw-data/workspace/projects/haritha-hospitals &&
  git checkout main &&
  git reset --hard <last-good-commit>'

docker exec erp-prod-backend-1 bash -c '
  cd /home/frappe/frappe-bench/apps/haritha_hospital &&
  git pull origin main'
```

```
# Reload code (no migrate — schema unchanged)
```

```
docker restart erp-prod-backend-1
```

No data loss. Code state reverts.

Level 3 — Restore from backup (last resort, destructive)

Use ONLY if Levels 1+2 fail AND user/data state is broken.

```
# 1. Stop bench to prevent writes during restore
docker exec erp-prod-backend-1 bash -c '
  supervisorctl stop frappe-bench-workers: frappe-bench-web: frappe-bench-schedule:'
```

```

# 2. Identify backup bundle (timestamp from before deploy)
ssh vijay@144.217.163.228 'ls -lt /home/vijay/backups/prod/' | head -5
# → pick the bundle from BEFORE the bad deploy, e.g.
#    pberpprod_backup_20260829_030000.tar.gz

# 3. Extract
ssh vijay@144.217.163.228 "cd /tmp && mkdir -p restore && cd restore && \
    tar -xzf /home/vijay/backups/prod/pberpprod_backup_20260829_030000.tar.gz"

# 4. Restore DB
docker cp /tmp/restore/*database.sql.gz erp-prod-backend-1:/tmp/
docker exec erp-prod-backend-1 bash -c '
    gunzip -c /tmp/*database.sql.gz | mysql -u root -p"${docker exec erp-prod-db-1 printenv
# 5. Restore site_config + files
docker cp /tmp/restore/*site_config_backup.json erp-prod-backend-1:/tmp/
docker exec erp-prod-backend-1 bash -c '
    cp /tmp/*site_config_backup.json /home/frappe/frappe-bench/sites/pberpprod.duckdns.org/s
docker cp /tmp/restore/*-files.tar erp-prod-backend-1:/tmp/
docker exec erp-prod-backend-1 bash -c '
    cd /home/frappe/frappe-bench/sites/pberpprod.duckdns.org &&
    rm -rf public/files private/files &&
    tar -xf /tmp/*-files.tar'

docker cp /tmp/restore/*-private-files.tar erp-prod-backend-1:/tmp/
docker exec erp-prod-backend-1 bash -c '
    cd /home/frappe/frappe-bench/sites/pberpprod.duckdns.org &&
    tar -xf /tmp/*-private-files.tar -C private/'

# 6. Restart everything
docker exec erp-prod-backend-1 bash -c '
    supervisorctl start frappe-bench-workers: frappe-bench-web: frappe-bench-schedule:'
docker restart erp-prod-backend-1
sleep 60

# 7. Verify
curl -sS -o /dev/null -w "%{http_code}\n" https://pberpPROD.duckdns.org/

Data state reverts to backup timestamp. Expect 15-30 min downtime. Communicate to Venkat
before proceeding.

```

Post-rollback

- ☐ Smoke test passes
- ☐ Notify Venkat: “prod rolled back to backup at {timestamp}”
- ☐ Investigate root cause (start post-mortem within 24h — see 05.2 Post-Mortem)
- ☐ Add new lesson to LEARNINGS.md if root cause is novel

Verification checklist (post-deploy)

[] curl https://pberpPROD.duckdns.org/ → 302 (redirect to login)

```
[ ] curl POST /api/method/login (Administrator)           → {"message":"Success"}
[ ] curl /api/resource/Company (with auth cookie)         → 200, list non-empty
[ ] Browser: login as Administrator → Roster loads (no 500)
[ ] Browser: HR → Attendance → today's record loads
[ ] Browser: HR → Shift Type list loads (CF "shift_code" visible if added)
[ ] tabError Log: SELECT COUNT(*) FROM `tabError Log`
    WHERE creation > DATE_SUB(NOW(), INTERVAL 5 MINUTE)    → 0 (or near-zero)
[ ] docker logs --tail 30 erp-prod-backend-1              → no ModuleNotFoundError
[ ] docker logs --tail 30 erp-prod-scheduler-1            → no tracebacks
[ ] Venkat ack received on Telegram
```

If any check fails → invoke 04.4 Incident Response SEV-2 or higher.

Emergency hotfix (skip QA)

For P0 bugs only — Venkat explicit approval required.

1. Branch from main: `git checkout -b hotfix/critical-bug`
2. Fix on dev, smoke-test
3. Get Venkat approval (Telegram)
4. Push to main
5. Pull on prod + restart (skip QA)
6. Smoke + ack to Venkat
7. Follow up with QA promote later (don't skip forever)

Communication

- **Before prod:** Telegram Venkat with commit + ETA
- **During deploy (if > 5 min):** status update every 5 min
- **After smoke passes:** “prod deploy complete. {commit} live. All smoke checks pass.”
- **If any check fails:** invoke 04.4 Incident Response — don't troubleshoot alone on prod

References

- **Architecture:** 00.2 Architecture Overview
- **Schema details:** 01.1 Schema Reference (274 customizations)
- **Migration script:** `projects/haritha-hospitals/scripts/migrate_master_data.py`
- **App layout:** 00.1 What is ERP / Frappe / HRMS / Custom App §“Custom App”
- **Lessons cited:** #46 (install-app restart), #48 (8-phase test), #72 (parent-verify), #79 (hardened backup), #151 (fixtures declaration), #152 (module column), #153 (gunicorn preload), #154 (docker restart vs down), #155 (Letter Head no module), #156 (PF/N stock seeds), #157 (master data upsert)

If this runbook conflicts with MEMORY.md, MEMORY.md wins. If MEMORY.md is wrong, fix MEMORY.md first.